

METIMMOX-1 Expected Value of Partially Perfect Information Analysis

Multi-Scenario Analysis for Biomarker-Guided Strategies in Metastatic MSS/pMMR Colorectal Cancer

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1 Summary

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

Table 1: Analysis scenario definitions. Higher WTP thresholds (€100,000 and €150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (€4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

Scenario Name	WTP Threshold	Nivolumab Cost
Base Case	€ 51,000	€13,923
Decreased Nivolumab Cost	€ 51,000	€ 4,641
WTP €100,000	€100,000	€13,923
WTP €100,000 + Decreased Nivolumab	€100,000	€ 4,641
WTP €150,000	€150,000	€13,923
WTP €150,000 + Decreased Nivolumab	€150,000	€ 4,641

2 Base Case Scenario

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €13,923

Total EVPI: €0

The EVPI is effectively zero, indicating no decision uncertainty at the base case WTP threshold. In this case, one strategy is clearly dominant regardless of parameter uncertainty, making EVPPI calculations less meaningful for the base case scenario.

2.1 Cost-Effectiveness Results

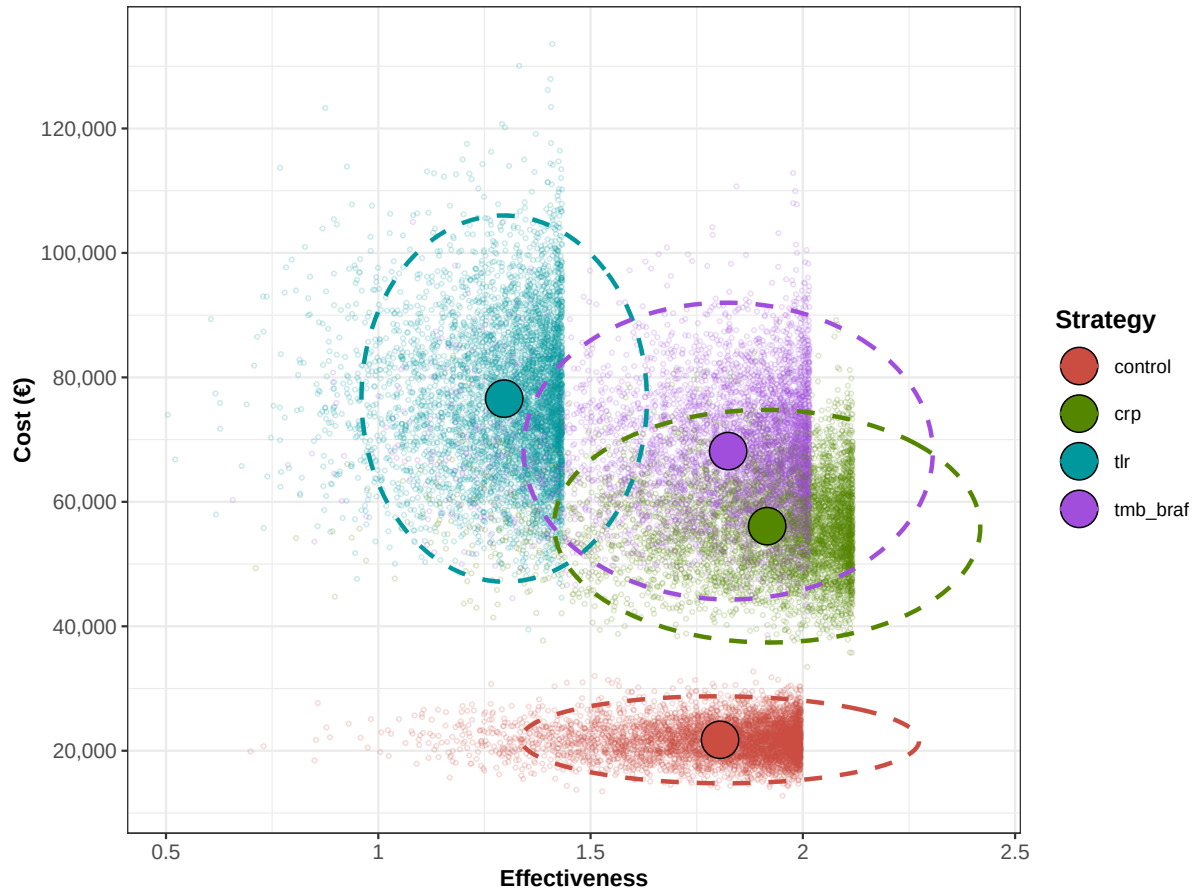


Figure 1: Cost-effectiveness scatter plot (base case)



Figure 2: Incremental cost-effectiveness scatter plot relative to control (base case)

Table 2: Cost-effectiveness analysis summary from probabilistic sensitivity analysis (base case)

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,758.31	1.805	€ 2,856.78	0.192
crp	€56,086.29	1.916	€ 7,638.79	0.205
tlr	€76,586.70	1.297	€12,026.29	0.137
tmb_braf	€68,154.60	1.824	€ 9,745.35	0.197

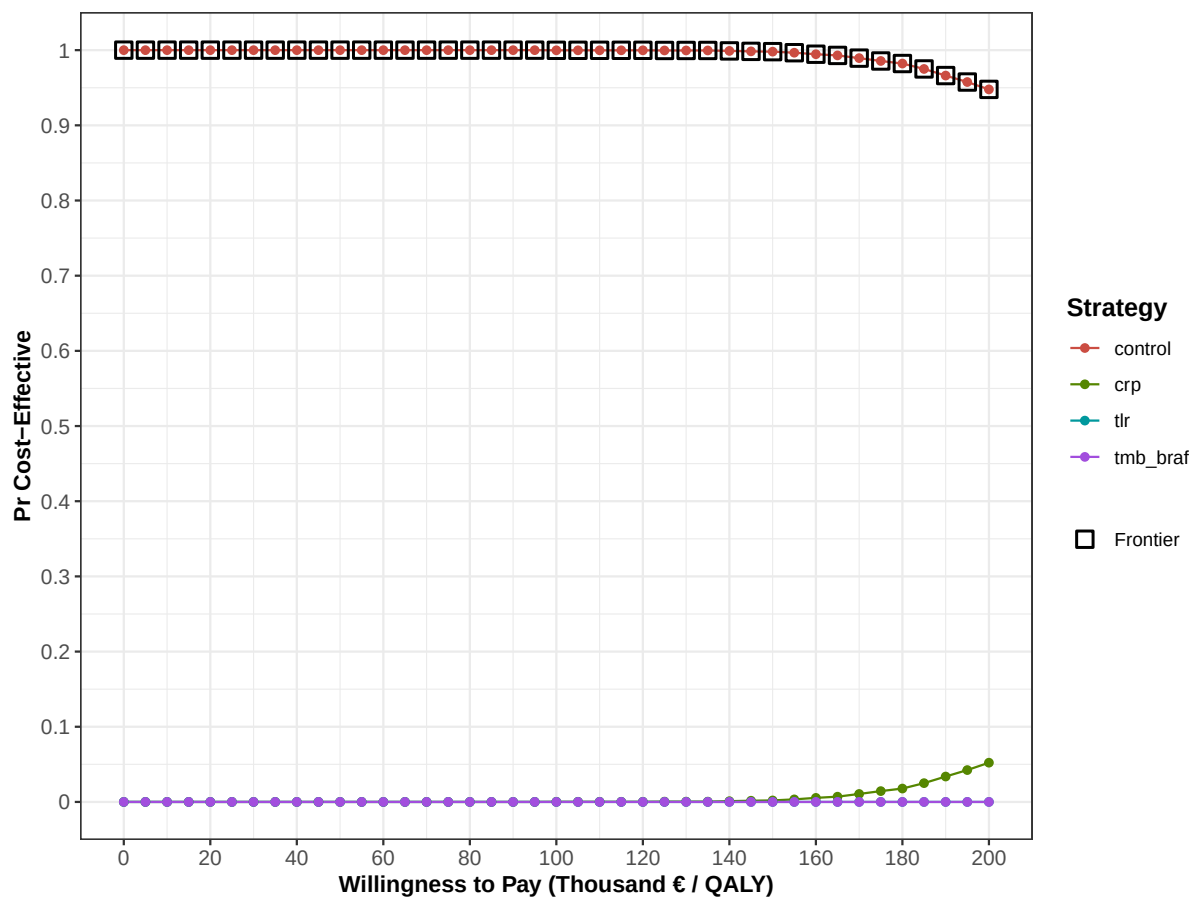


Figure 3: Cost-effectiveness acceptability curves (base case)

2.2 EVPPI Results

3 Scenario 1: Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €4,641 (80% discount)

Total EVPI: €4.01

Reducing nivolumab costs creates decision uncertainty, making EVPPI analysis meaningful.

3.1 Cost-Effectiveness Results

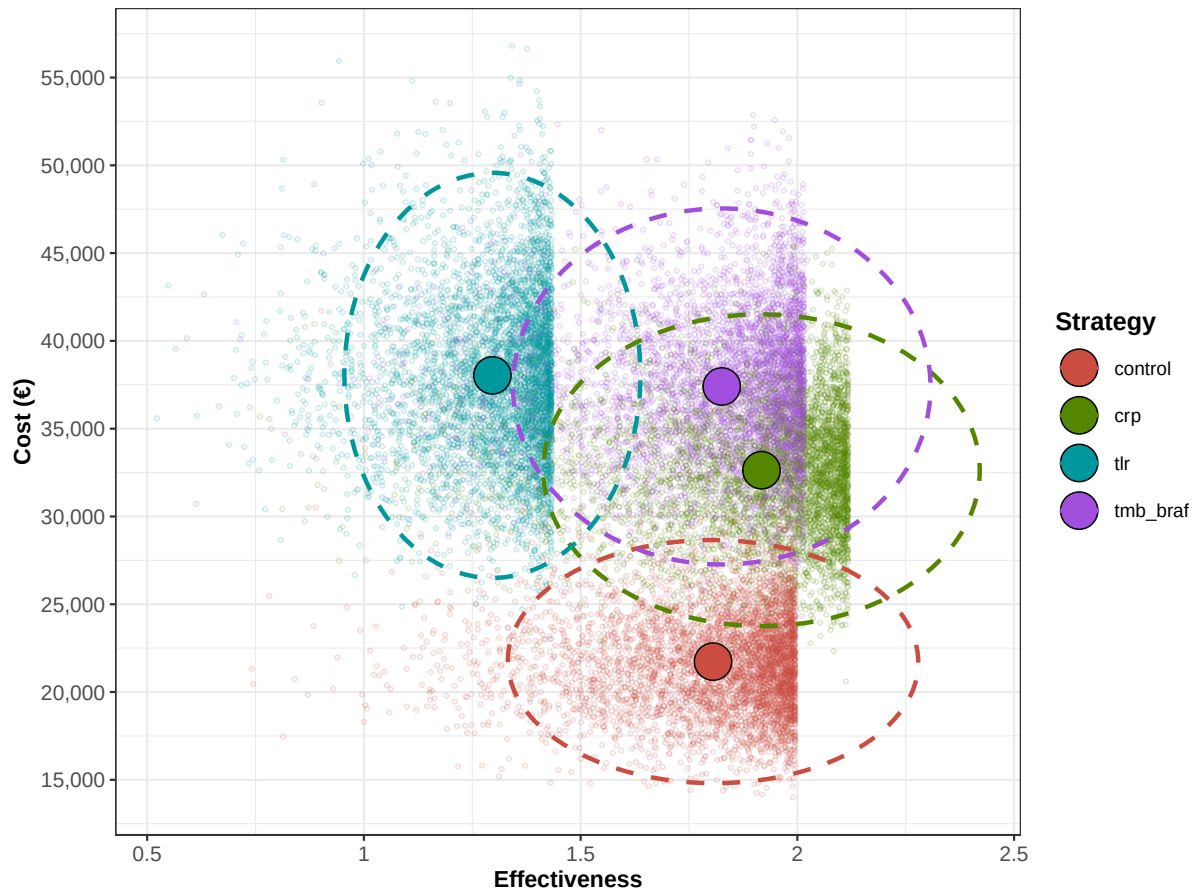


Figure 4: Cost-effectiveness scatter plot (Scenario 1)



Figure 5: Incremental cost-effectiveness scatter plot relative to control (Scenario 1)

Table 3: Cost-effectiveness analysis summary (Scenario 1)

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,731.75	1.806	€2,829.06	0.193
crp	€32,633.36	1.917	€3,627.37	0.206
tlr	€38,034.36	1.296	€4,713.70	0.140
tmb_braf	€37,404.23	1.826	€4,142.88	0.197

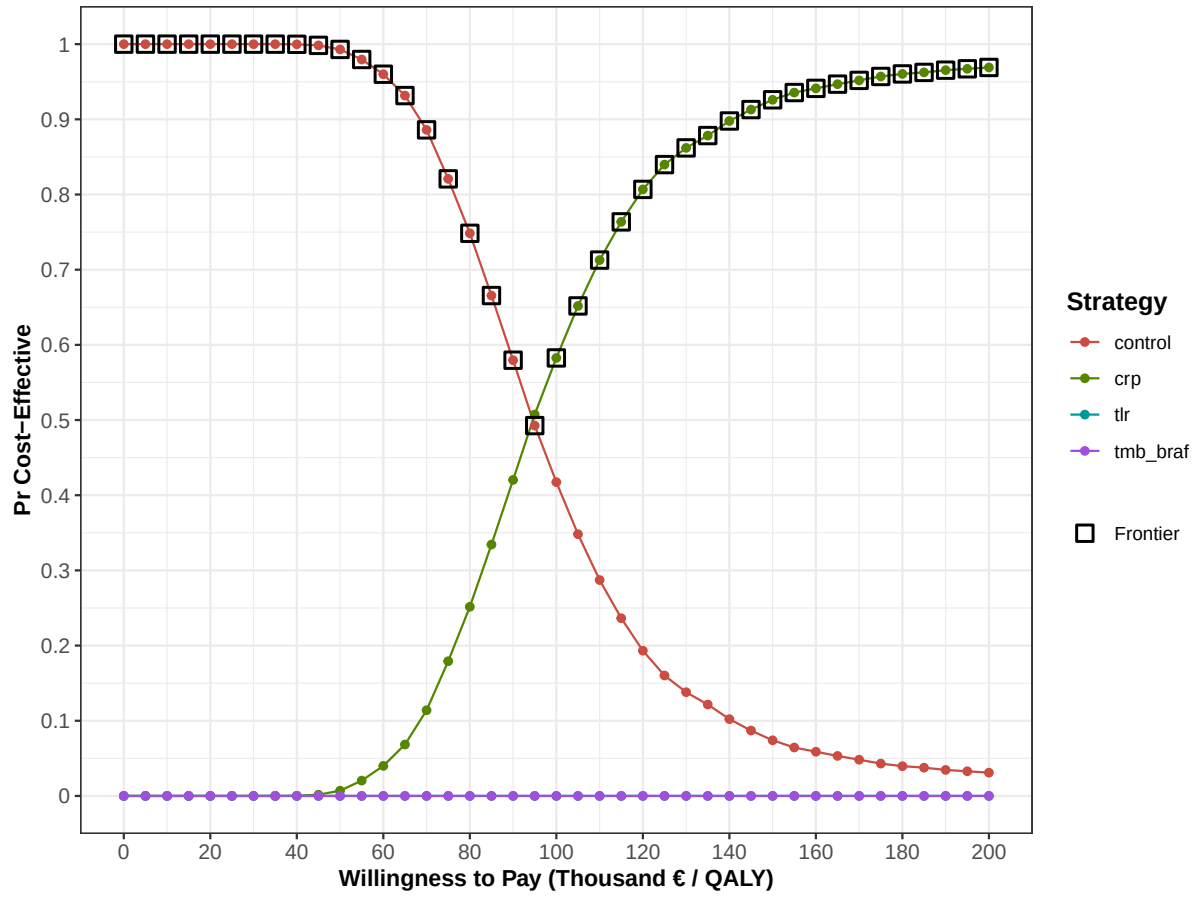


Figure 6: Cost-effectiveness acceptability curves (Scenario 1)

3.2 EVPPI Results

Table 4: EVPPI results: Decreased nivolumab cost

	Parameter	EVPPI	% of EVPI
1	c_drug_nivo	€4.01	100%
4	c_test_CT	€4.01	100%
5	u_np	€4.01	100%
6	u_p	€4.01	100%
7	c_other_last	€4.01	100%
9	c_other_baseline	€4.01	100%
2	c_drug_FLOX	€0	0%
3	c_test_NGS	€0	0%
8	c_other_visit	€0	0%
10	c_other_follow	€0	0%

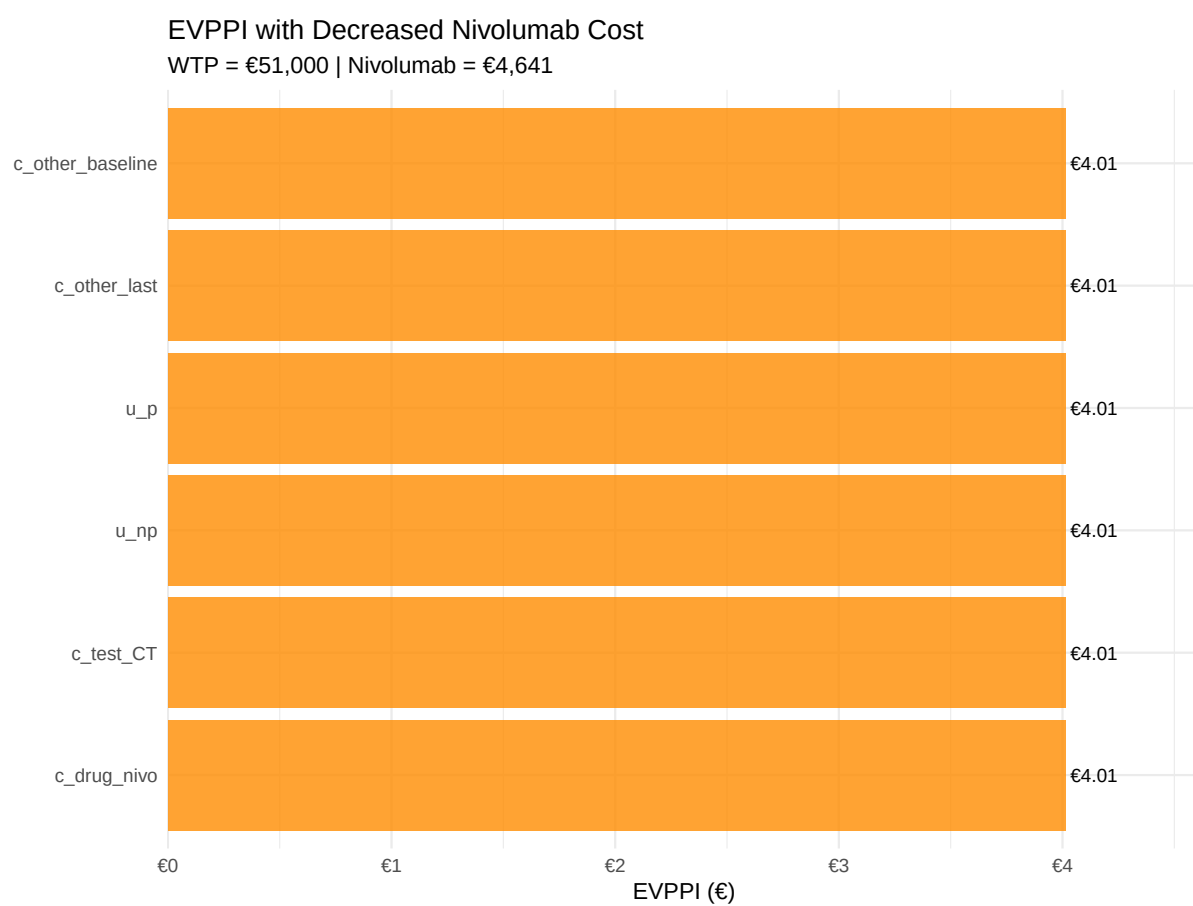


Figure 7: EVPPPI with decreased nivolumab cost

4 Scenario 2: Higher WTP Threshold (€100,000)

Willingness-to-Pay Threshold: €100,000

Nivolumab Cost: Standard (base case)

Total EVPI: €0

4.1 Cost-Effectiveness Results

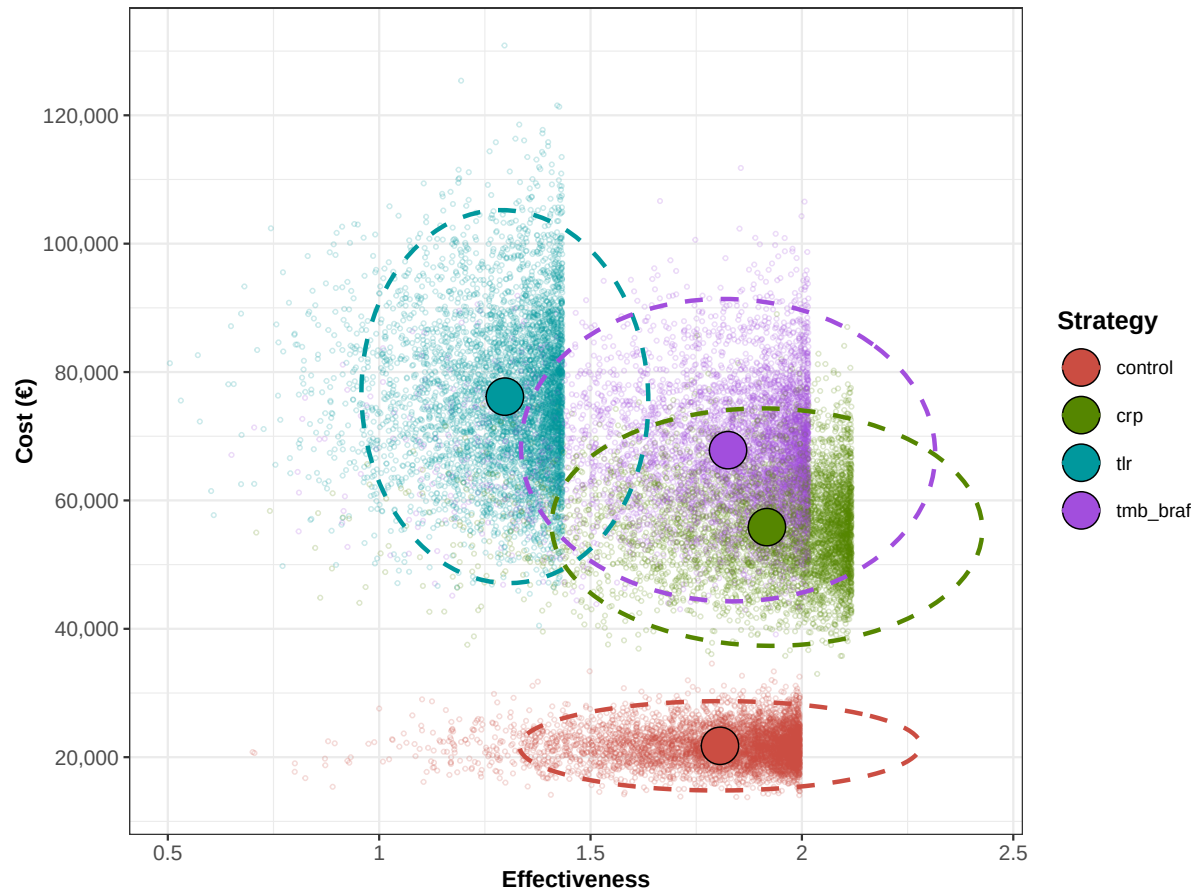


Figure 8: Cost-effectiveness scatter plot (Scenario 2)

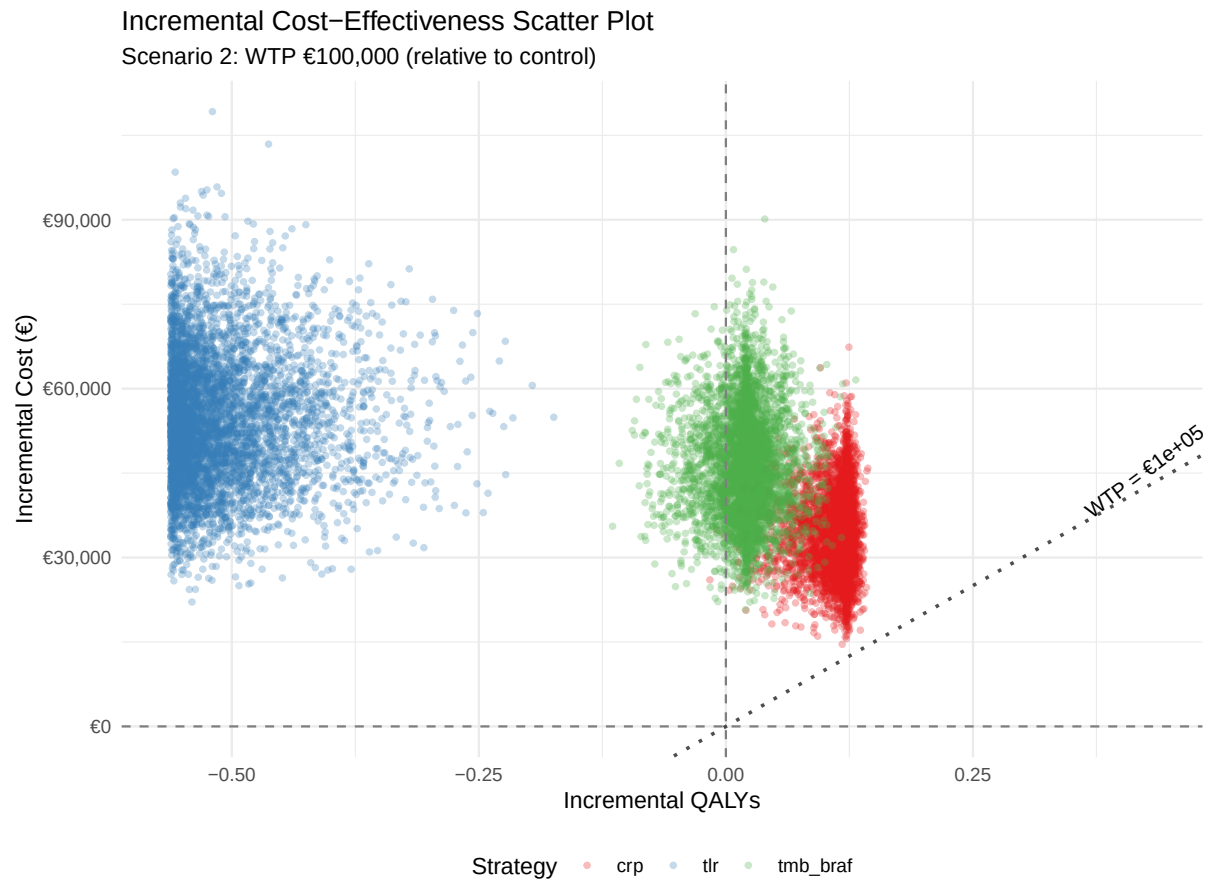


Figure 9: Incremental cost-effectiveness scatter plot relative to control (Scenario 2)

Table 5: Cost-effectiveness analysis summary (Scenario 2)

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,772.44	1.806	€ 2,843.93	0.194
crp	€55,841.03	1.917	€ 7,557.27	0.208
tlr	€76,171.90	1.297	€11,872.79	0.139
tmb_braf	€67,830.11	1.825	€ 9,626.76	0.200

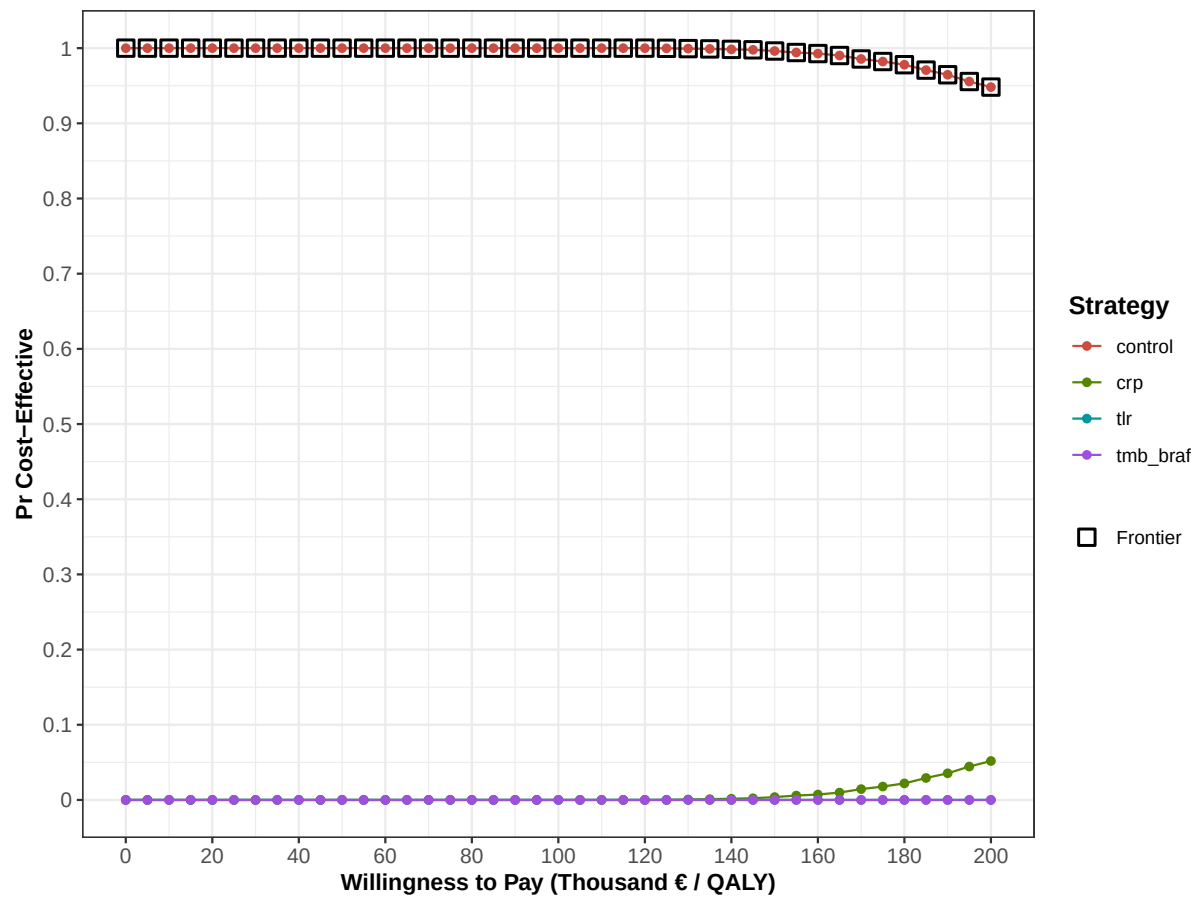


Figure 10: Cost-effectiveness acceptability curves (Scenario 2)

4.2 EVPPI Results

5 Scenario 3: Higher WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €100,000

Nivolumab Cost: €4,641 (80% discount)

Total EVPI: €1081.33

5.1 Cost-Effectiveness Results

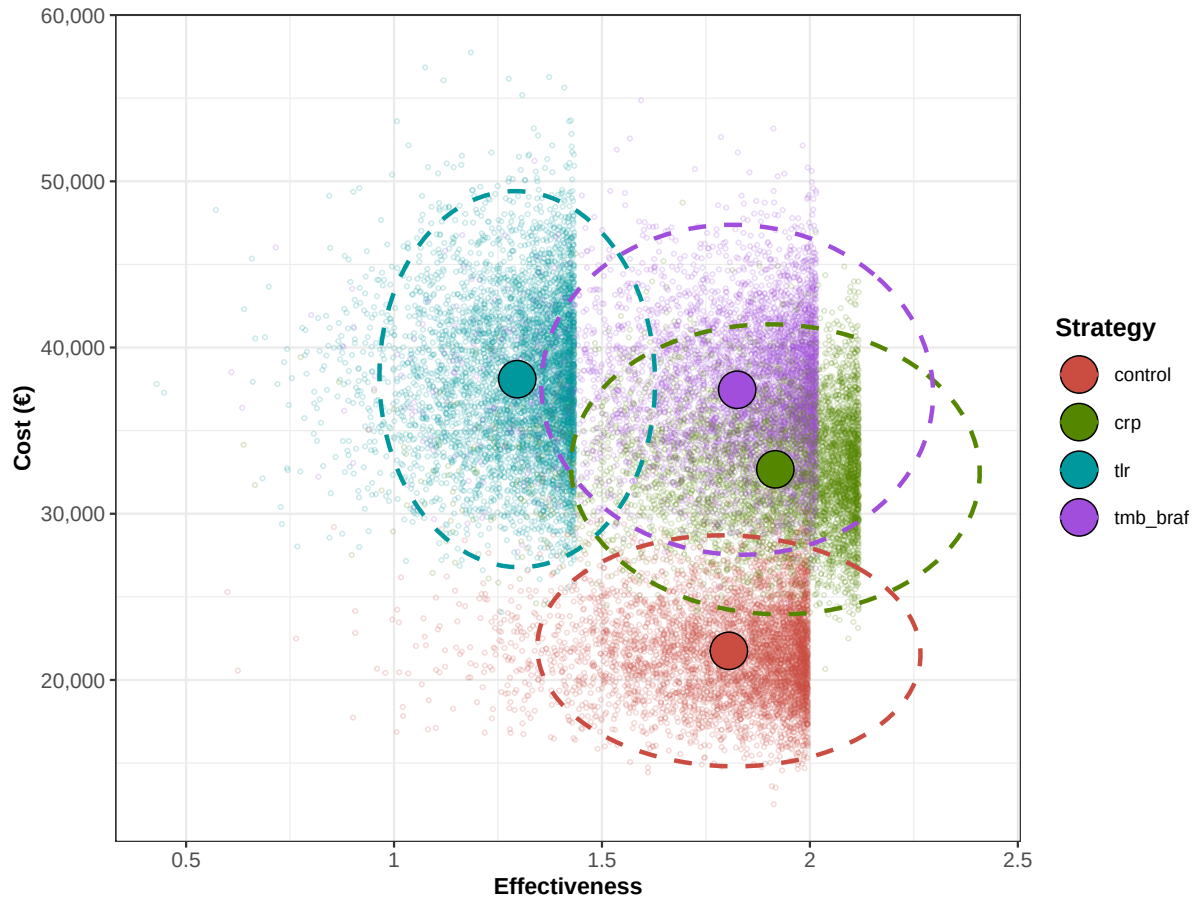


Figure 11: Cost-effectiveness scatter plot (Scenario 3)

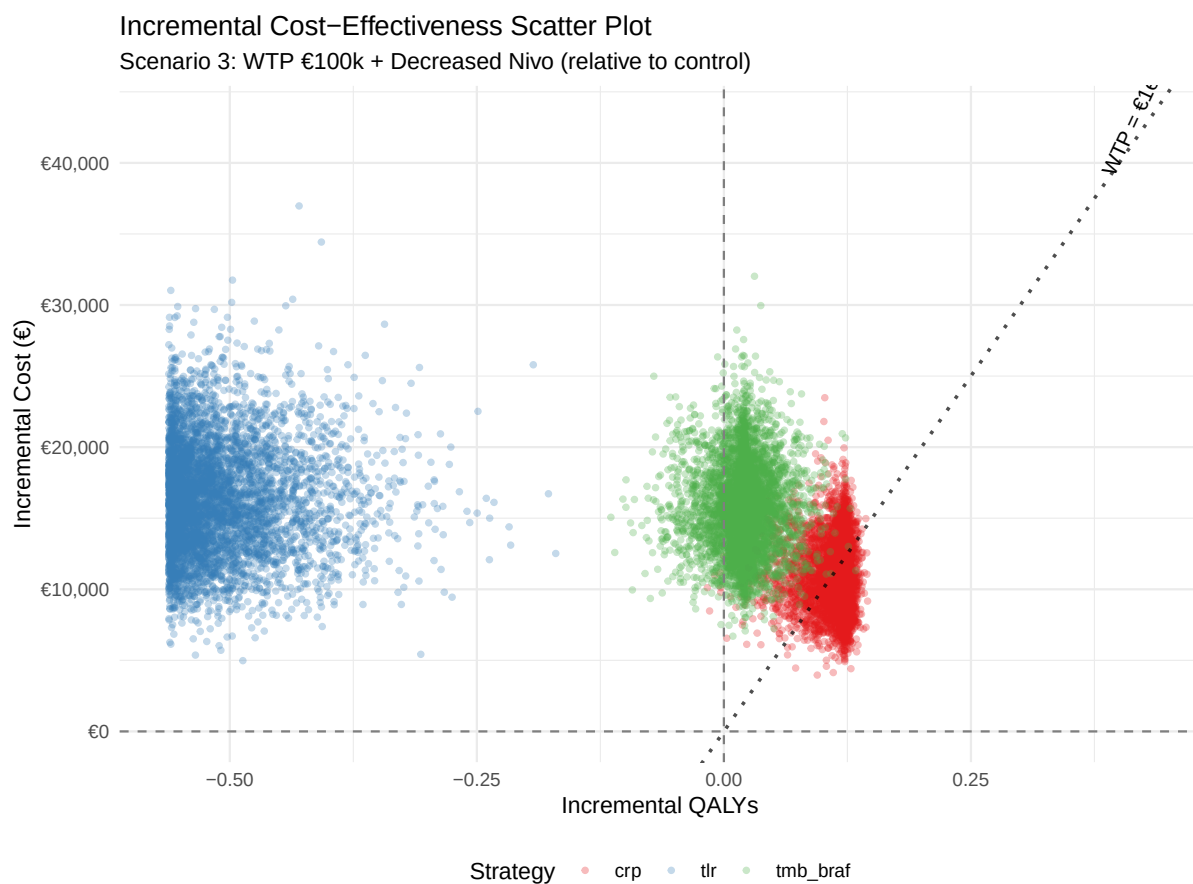


Figure 12: Incremental cost-effectiveness scatter plot relative to control (Scenario 3)

Table 6: Cost-effectiveness analysis summary (Scenario 3)

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,753.79	1.805	€2,839.77	0.188
crp	€32,673.82	1.917	€3,564.38	0.201
tlr	€38,096.58	1.296	€4,622.78	0.135
tmb_braf	€37,458.96	1.825	€4,055.46	0.192

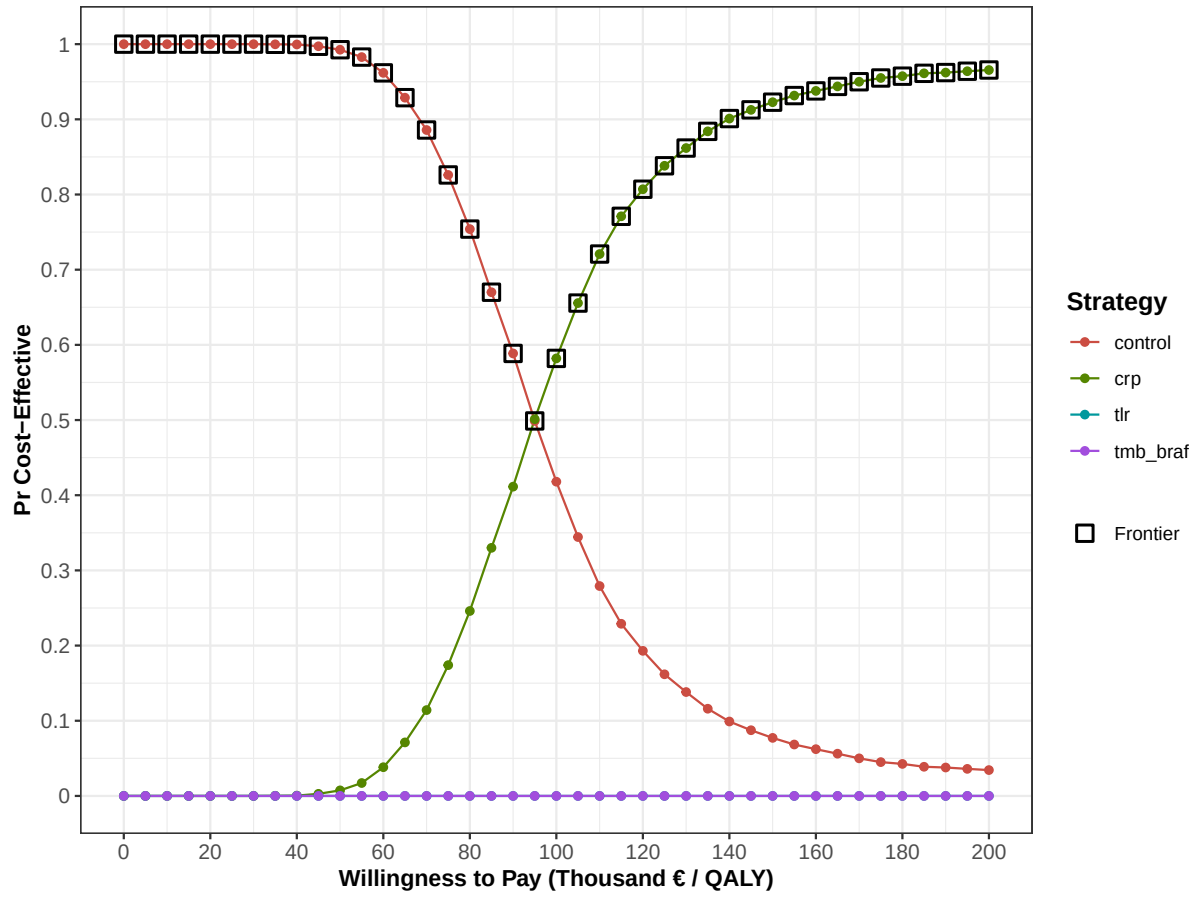


Figure 13: Cost-effectiveness acceptability curves (Scenario 3)

5.2 EVPPI Results

Table 7: EVPPI results: WTP €100,000 + decreased nivolumab

	Parameter	EVPPI	% of EVPI
4	c_test_CT	€1081.33	100%
5	u_np	€1081.33	100%
9	c_other_baseline	€1081.33	100%
8	c_other_visit	€942.17	87.1%
6	u_p	€736.93	68.1%
1	c_drug_nivo	€521.3	48.2%
10	c_other_follow	€16.59	1.5%
2	c_drug_FLOX	€0	0%
3	c_test_NGS	€0	0%
7	c_other_last	€0	0%

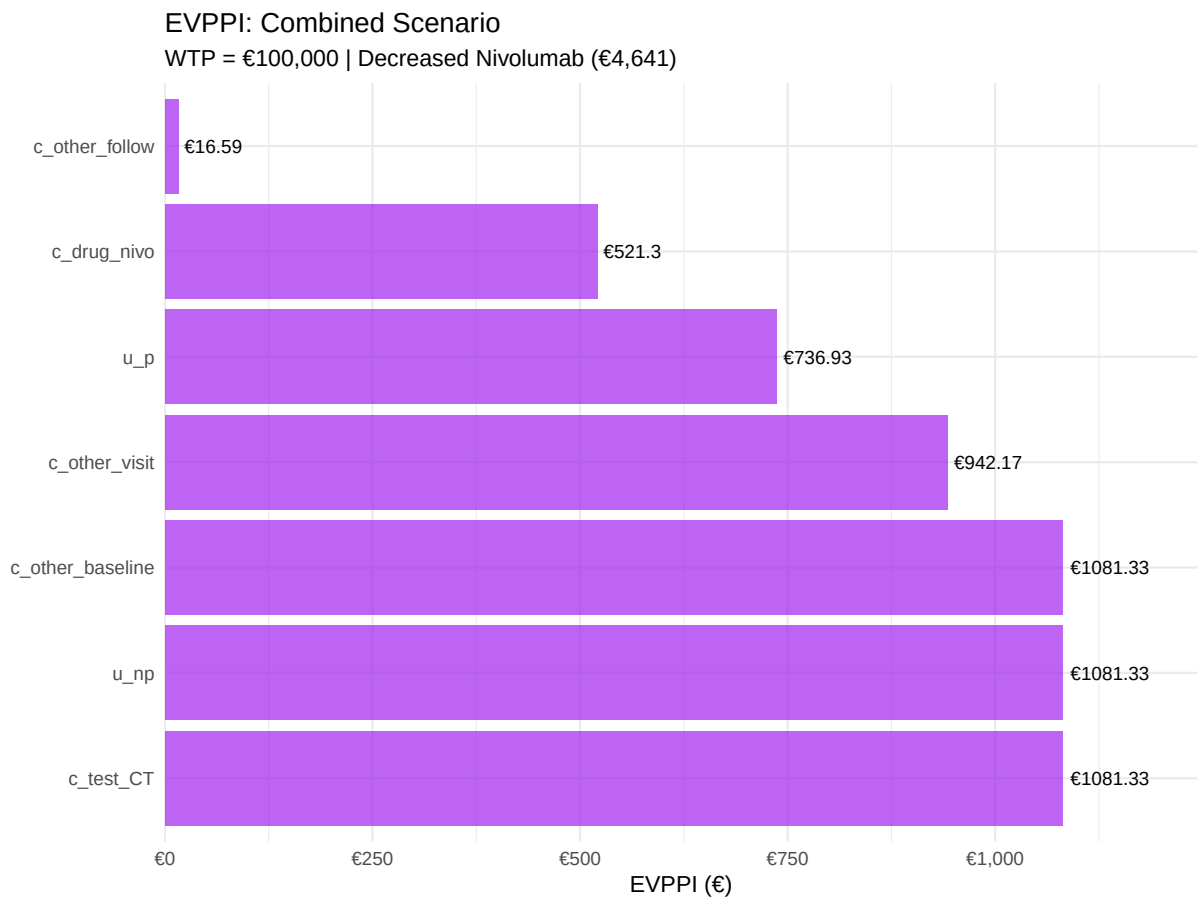


Figure 14: EVPPi: WTP €100,000 with decreased nivolumab

6 Scenario 4: Highest WTP Threshold (€150,000)

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: Standard (base case)

Total EVPI: €6.02

6.1 Cost-Effectiveness Results

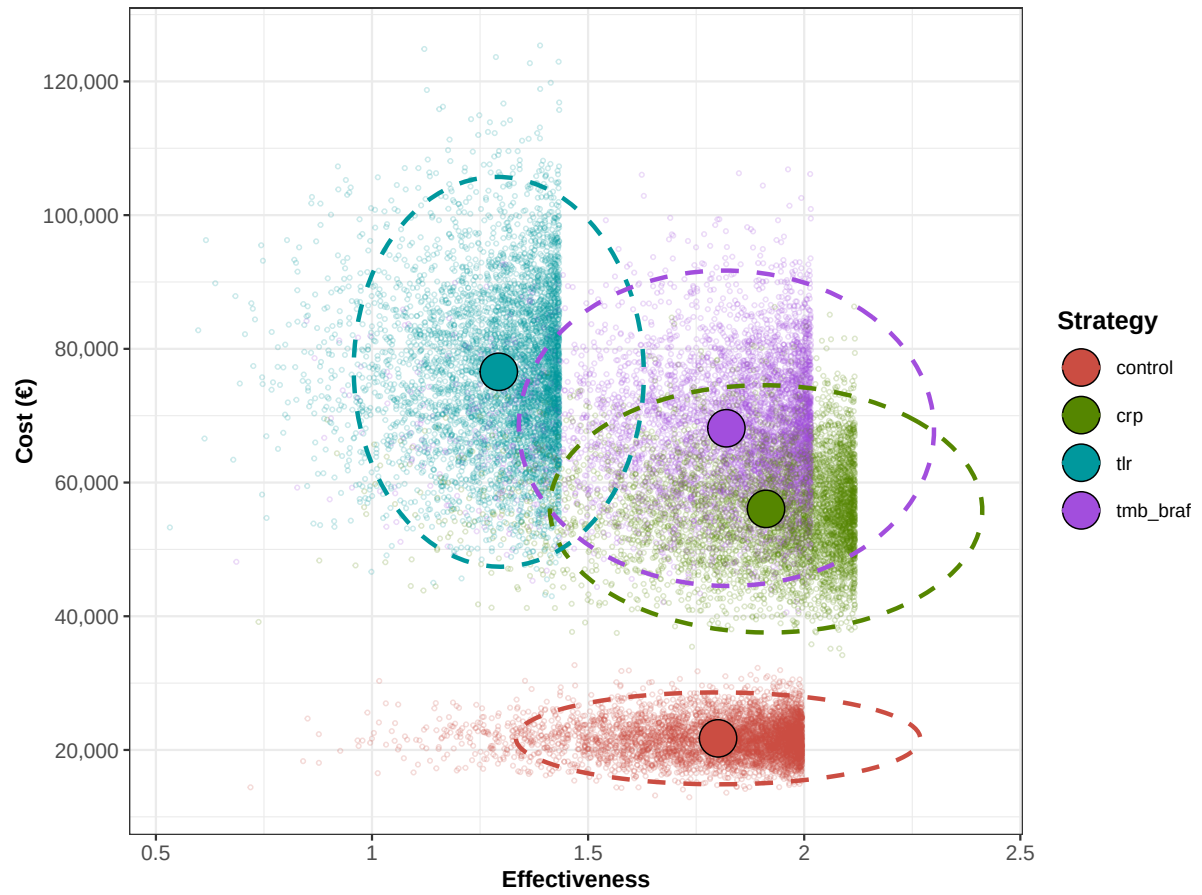


Figure 15: Cost-effectiveness scatter plot (Scenario 4)



Figure 16: Incremental cost-effectiveness scatter plot relative to control (Scenario 4)

Table 8: Cost-effectiveness analysis summary (Scenario 4)

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,734.74	1.801	€ 2,807.55	0.191
crp	€56,060.40	1.912	€ 7,557.54	0.204
tlr	€76,564.22	1.293	€11,912.95	0.137
tmb_braf	€68,120.65	1.820	€ 9,638.89	0.196

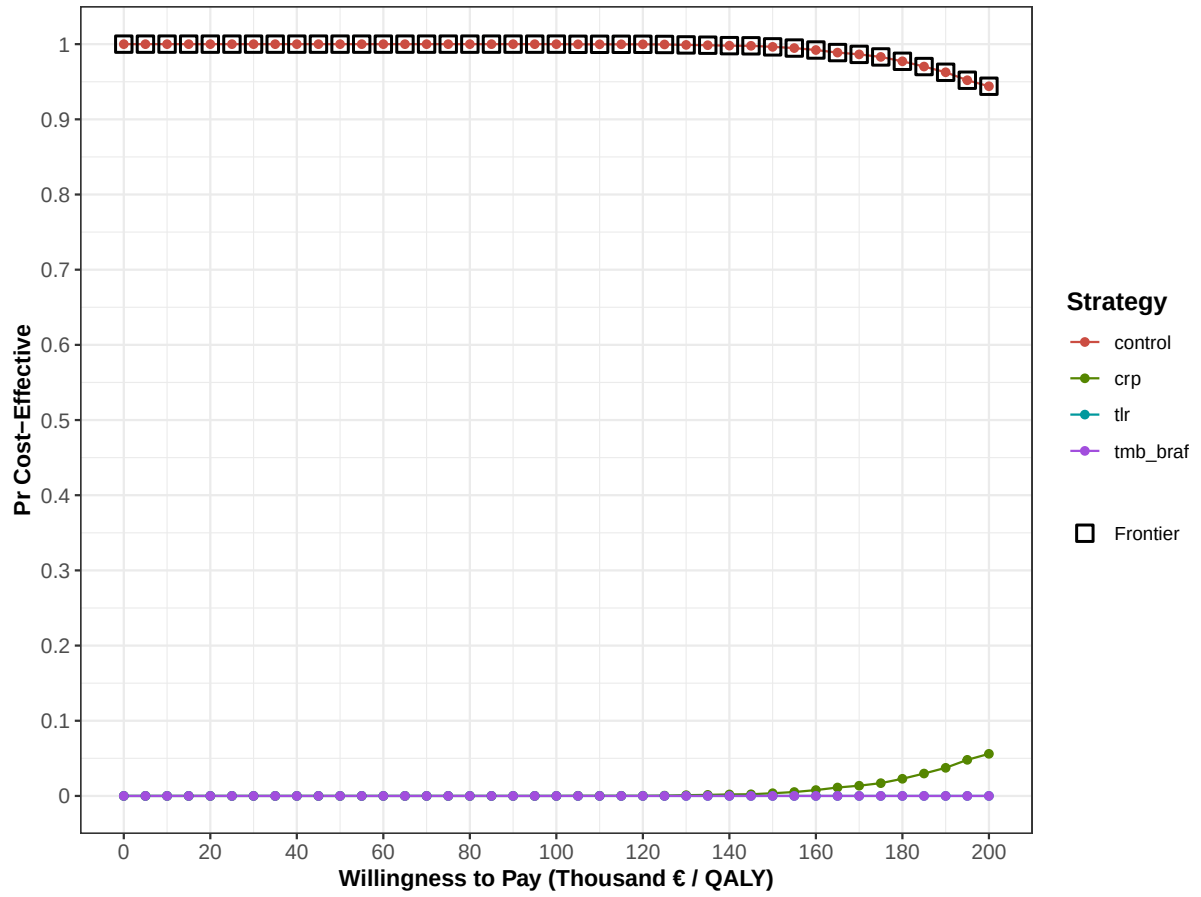


Figure 17: Cost-effectiveness acceptability curves (Scenario 4)

6.2 EVPPI Results

Table 9: EVPPI results: WTP €150,000

	Parameter	EVPPI	% of EVPI
1	c_drug_nivo	€6.02	100%
3	c_test_NGS	€6.02	100%
4	c_test_CT	€6.02	100%
5	u_np	€6.02	100%
6	u_p	€6.02	100%
7	c_other_last	€6.02	100%
2	c_drug_FLOX	€0	0%
8	c_other_visit	€0	0%
9	c_other_baseline	€0	0%
10	c_other_follow	€0	0%

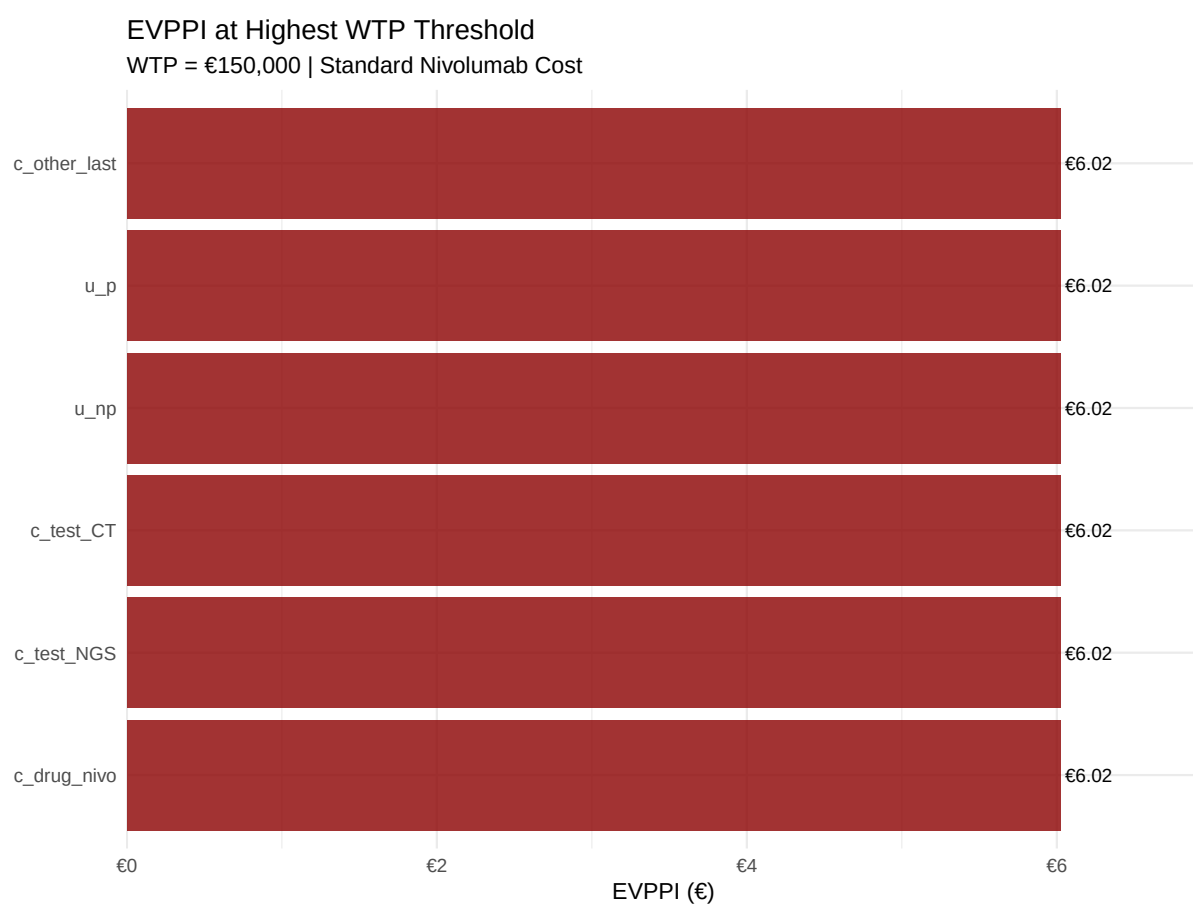


Figure 18: EVPPPI at WTP €150,000

7 Scenario 5: Highest WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: €4,641 (80% discount)

Total EVPI: €273.11

7.1 Cost-Effectiveness Results

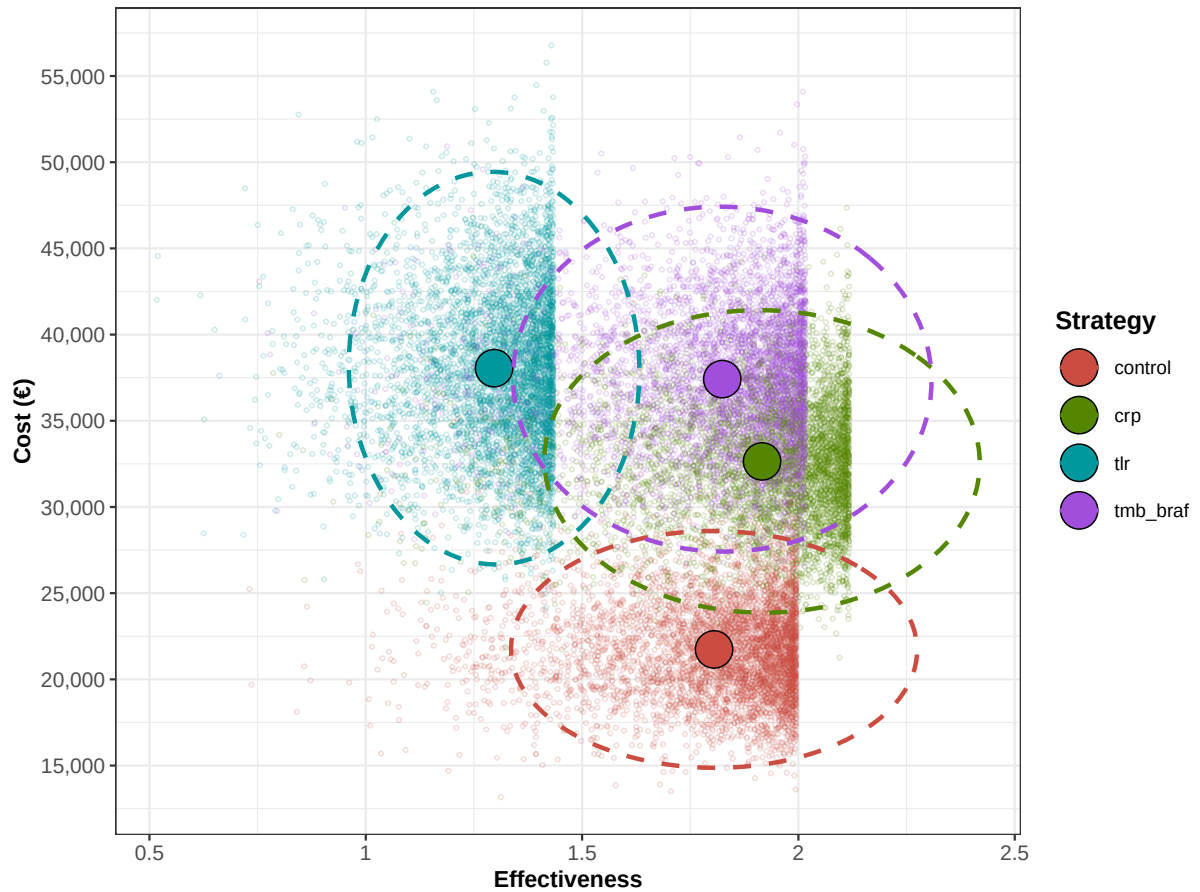


Figure 19: Cost-effectiveness scatter plot (Scenario 5)

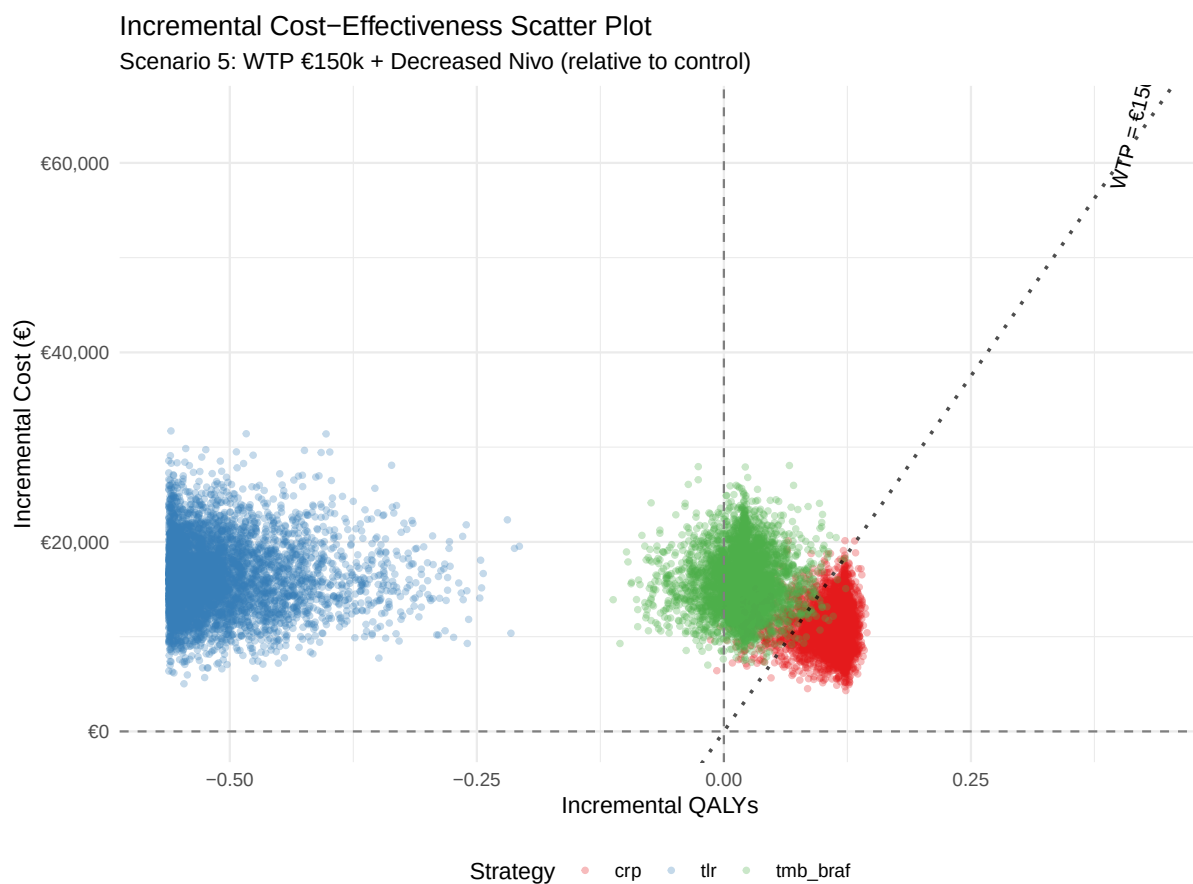


Figure 20: Incremental cost-effectiveness scatter plot relative to control (Scenario 5)

Table 10: Cost-effectiveness analysis summary (Scenario 5)

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,736.64	1.805	€2,806.95	0.192
crp	€32,643.39	1.916	€3,585.45	0.206
tlr	€38,052.81	1.297	€4,653.81	0.137
tmb_braf	€37,416.03	1.824	€4,088.11	0.198

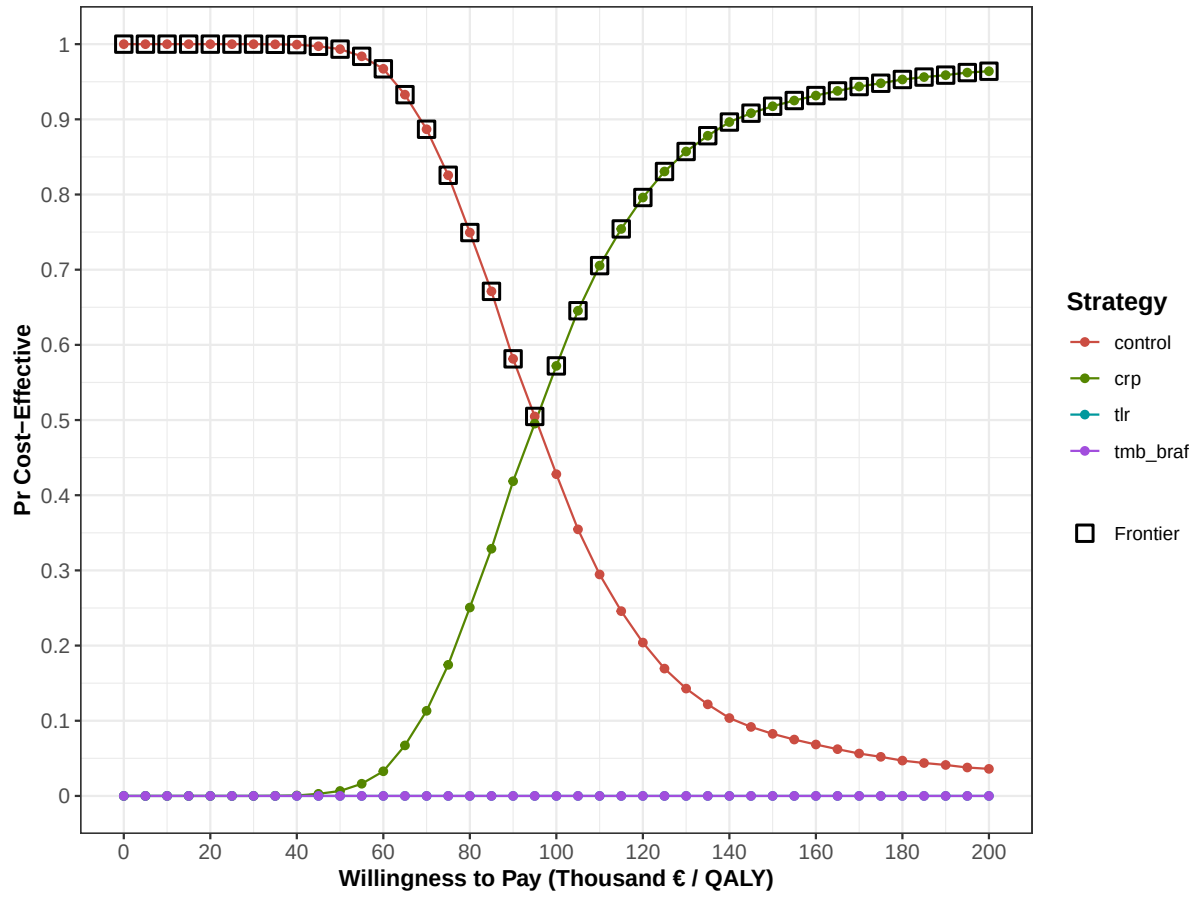


Figure 21: Cost-effectiveness acceptability curves (Scenario 5)

7.2 EVPPI Results

Table 11: EVPPI results: WTP €150,000 + decreased nivolumab

	Parameter	EVPPI	% of EVPI
1	c_drug_nivo	€273.11	100%
2	c_drug_FLOX	€273.11	100%
3	c_test_NGS	€273.11	100%
5	u_np	€273.11	100%
6	u_p	€273.11	100%
7	c_other_last	€273.11	100%
9	c_other_baseline	€273.11	100%
4	c_test_CT	€0	0%
8	c_other_visit	€0	0%
10	c_other_follow	€0	0%

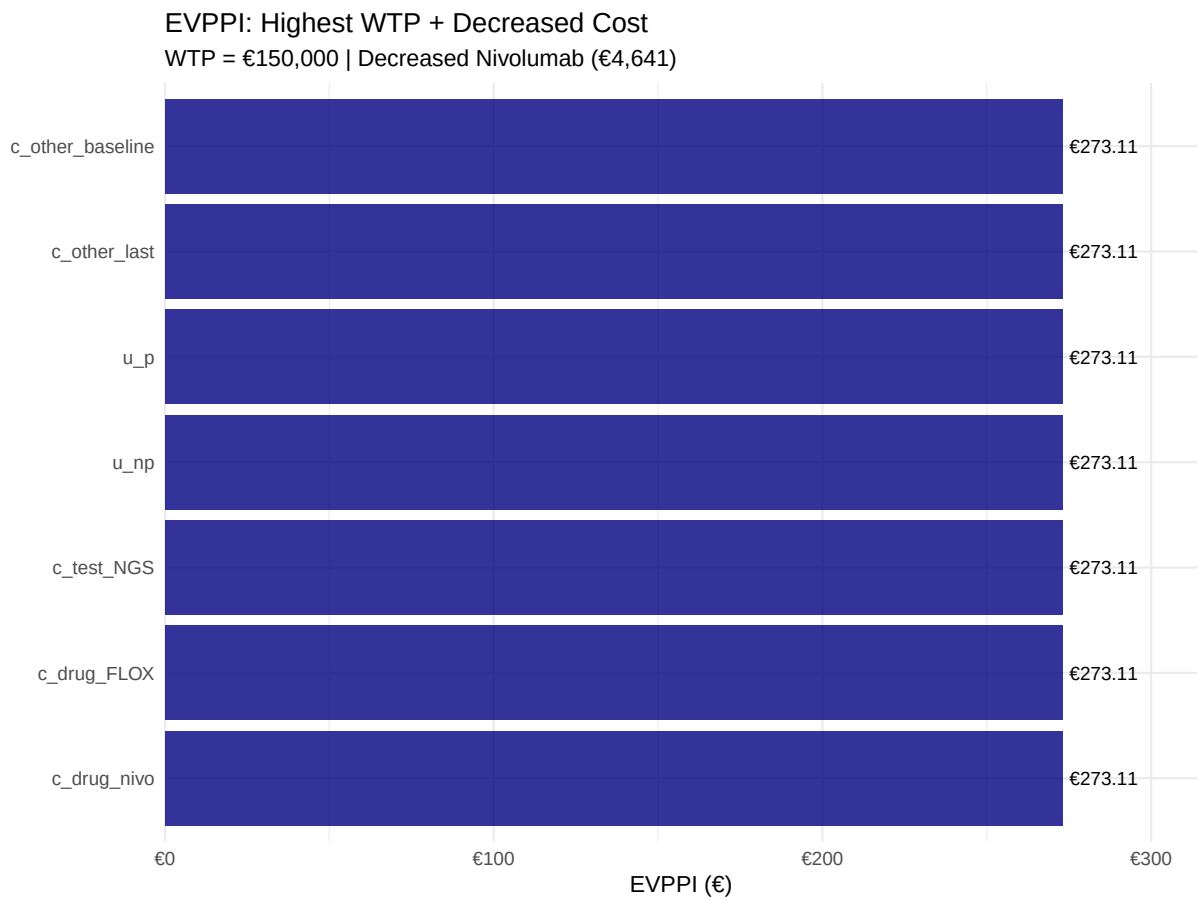


Figure 22: EVPPPI: Highest WTP with decreased nivolumab cost

8 Comparative Analysis Across Scenarios

Table 12: Summary of EVPI and top EVPPI by scenario

Scenario	WTP	EVPI	Top Parameter	Max EVPPI	% of EVPI
Decreased Nivolumab Cost	€ 51,000	€4.01	c_drug_nivo	€4.01	100
WTP €100,000 + Decreased Nivolumab	€100,000	€1081.33	c_test_CT	€1081.33	100
WTP €150,000	€150,000	€6.02	c_drug_nivo	€6.02	100
WTP €150,000 + Decreased Nivolumab	€150,000	€273.11	c_drug_nivo	€273.11	100

8.1 EVPI Comparison



Figure 23: Expected Value of Perfect Information across scenarios

8.2 Parameter Importance Heatmap

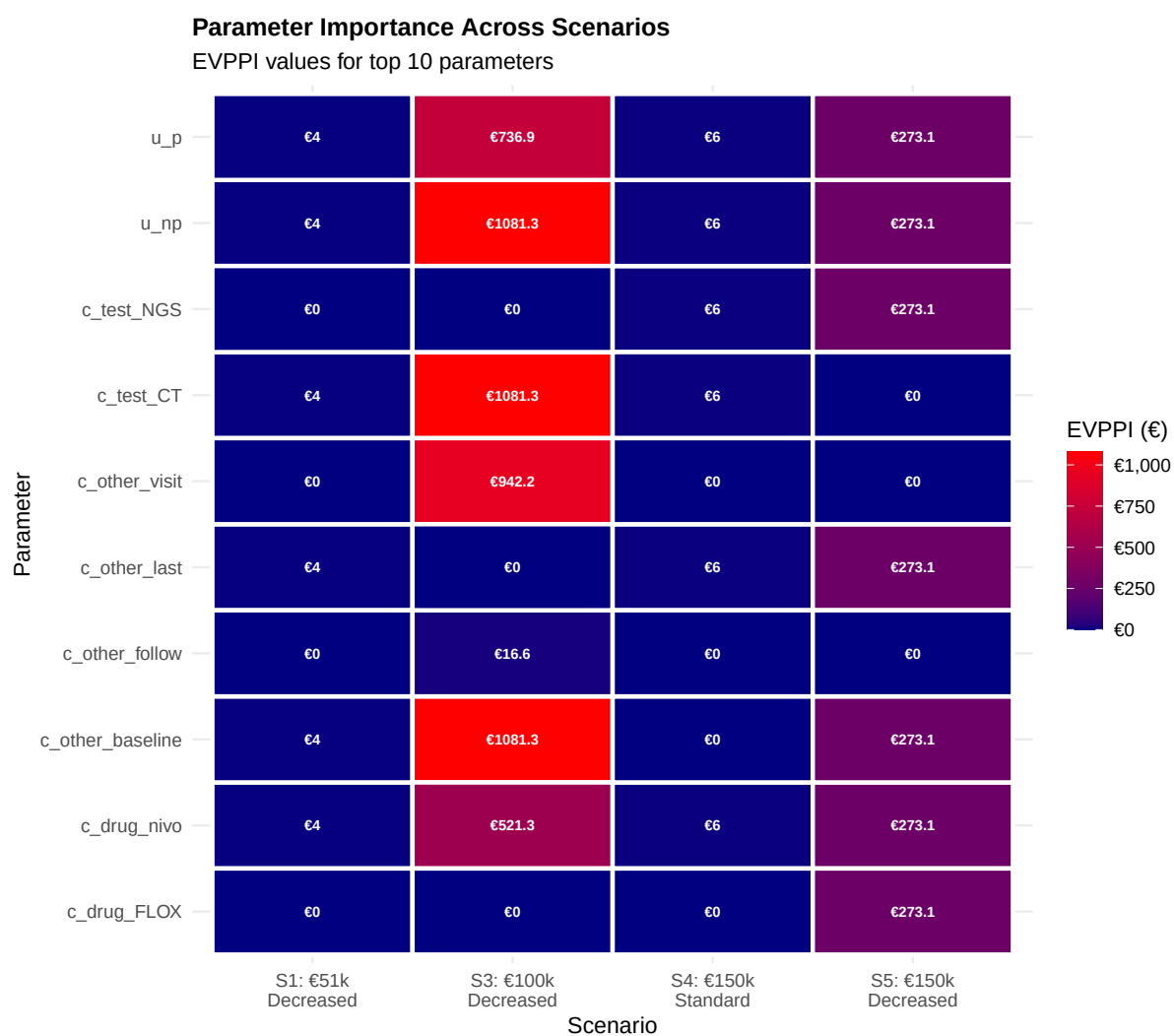


Figure 24: Parameter importance across scenarios (top 10 parameters)

9 Conclusions and Recommendations

9.1 Summary of Findings

Scenario with Greatest Decision Uncertainty: WTP €100,000 + Decreased Nivolumab (EVPI = €1081.33)

Most Important Parameter Across All Scenarios: u_{np} (Total EVPPI = €1364.48)

Impact of WTP Threshold: Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

Impact of Nivolumab Cost: Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.

9.2 Research Priorities

Based on the EVPPI analysis, future research should prioritize:

Table 13: Top 5 research priorities based on EVPPI

Priority	Parameter	Max EVPPI	Avg EVPPI	Relevant Scenarios
1	c_other_baseline	€1081.33	€339.61	3
2	c_test_CT	€1081.33	€272.84	3
3	u_{np}	€1081.33	€341.12	4
4	c_other_visit	€942.17	€235.54	1
5	u_p	€736.93	€255.02	4

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